

A Packing of 7 Four-Spaces in $\mathbb{R}^9$

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October 9, 2026

Abstract

We give 7 4-dimensional subspaces of $\mathbb{R}^9$ whose smallest squared chordal distance is at least 2.592416. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.539492, obtained by deleting 1 of the 8 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 7 subspaces of dimension 4 in $\mathbb{R}^9$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.592416 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 9$, $n = 4$. The cell $N = 7$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.539492, obtained by deleting 1 of the 8 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(9 - 4)/9 \cdot 7/6 = 2.592593$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = Y G^{-1} Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m9n4N7.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

152495074577729898287251941683885883043297398371518688249119040208718687
541349416080455979363035404594895795925860708415233160480195982415545056
414410386603035255709972618881780986291391417439

denominator:

588235294118015152499668954092245849619599206310288471526946974078297692
960682763026094051134628086445290282093515194461779748069615887551734699
98784105895945286127873712334143299049979416284

B Coordinates

7 blocks of 9 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m9n4N7.txt`.

51721737412 -399598378498 326685702368 356978980751295427225957 -785984486125 340380638991 -3150179218065544179840 589867696906 470433035690 -249258407610
188767471358 44273531811 -104257625686 -6831289441679283332247 -49655652486 317397398650 443308524356196490567373 161028193514 5730687449 111917622607
-117316856298 148187756040 -64267796056 -32889662325647539310411 -410626038607 -163429990238 6775698790641458429 504117335140 -57320885594 380055342739
610154591561 155939007351 285802406403 -45797571857-36960943879 193603634040 211954163307 16854630871059697009909 -29970738669 -249290938018 -42003682909
103732807251 -780779538868 -102264785647 -2689362559469242341 190124000266 215720827477 44530552562 -63674521094 689699332337 -473475062282 510905058158
232008099654 -299209012157 413201314939 -877685598851476093754 -69295261388 -523326700284 140070005295209934063782 311539373077 114425391536 -449131069900
128059373518 284003947967 506558629082 -53142457691587116265586 104453017352 383521090714 35253376862582369885835 -345090573082 -177581818547 184729330348
-637151191193 -25892499180 582342797547 11523583540925404896343 197163697006 -315716151845 7456916207101032422235 -71878998157 112437179929 318428447844
297809899218 107763722942 140268466292 699284491450189681512343 13977756793 738710387028 198761979683 90851633475 451996348347 95780435070 -472084832396
266670230285 615615261823 -124735779699 -208419685407110665626 -329112386439 355951809271 -13628282689420888951871 163263263740 332155711557 250677503991
33877864397 174835962801 558168671753 179190210074 283337571602 -532752364645 -2531810801 26715576782812675194022 -258167250157 -616830479385 -238129309755
-735270769669 423570601371 -313892475931 -234063515746364036329 -236741965737 -233788896139 -2414912420717917213 -67080470734 395504302028 239921912682
-62897487109 277700349351 237339145666 185412972772-240127539620 451228914917 159357304186 -35276254874457671100217 605970451027 677841557 -197046383788
270467643629 378383728813 134064564985 -34830954895287452571193 182379828830 -26095958347 -70427653731-111031443259 -127823762261 -212939164594 518982989416
199303970788 -353786463573 -301329829776 -5847261634604528169738 -514746468817 -10037457432 -49929153350664807023 -536479042854 -48893417577 -162964736513
204517088681 143129483211 -75814299681 -86638794633-488886026207 -256325361487 414105026386 -2650988130524415482 119266918504 -244412642407 -298868384533
-98607353421 -98400747923 5954229945737 -5903114162472663012227 -126307033006 354720930837 -1693426181656369697917 -503542278806 140160979975 32635467655
-467690721778 -188182977321 221297899698 -12684765070588834866 -308947223783 343994454236 -17931604097077992341413 192749651110 -371336348776 578389646304
-71898869360 112334326342 -448830425142 -54003384750242078935 36466208712 -420573691055 -607832216691-632975776520 73066695220 655737131189 24899027829
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-103109812891 -311248815735 -420594953458 590015601055751511191 59814352923 -413328499853 -37549650807-441517512808 -70332615028 -90885298231 205221347902